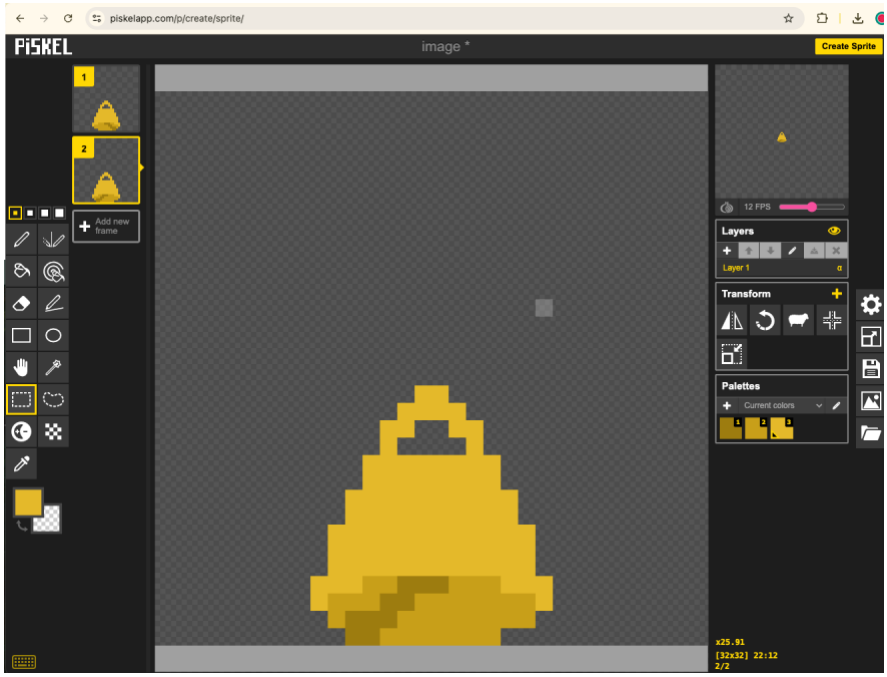
Process & Decision Documentation

Project/Assignment Decisions

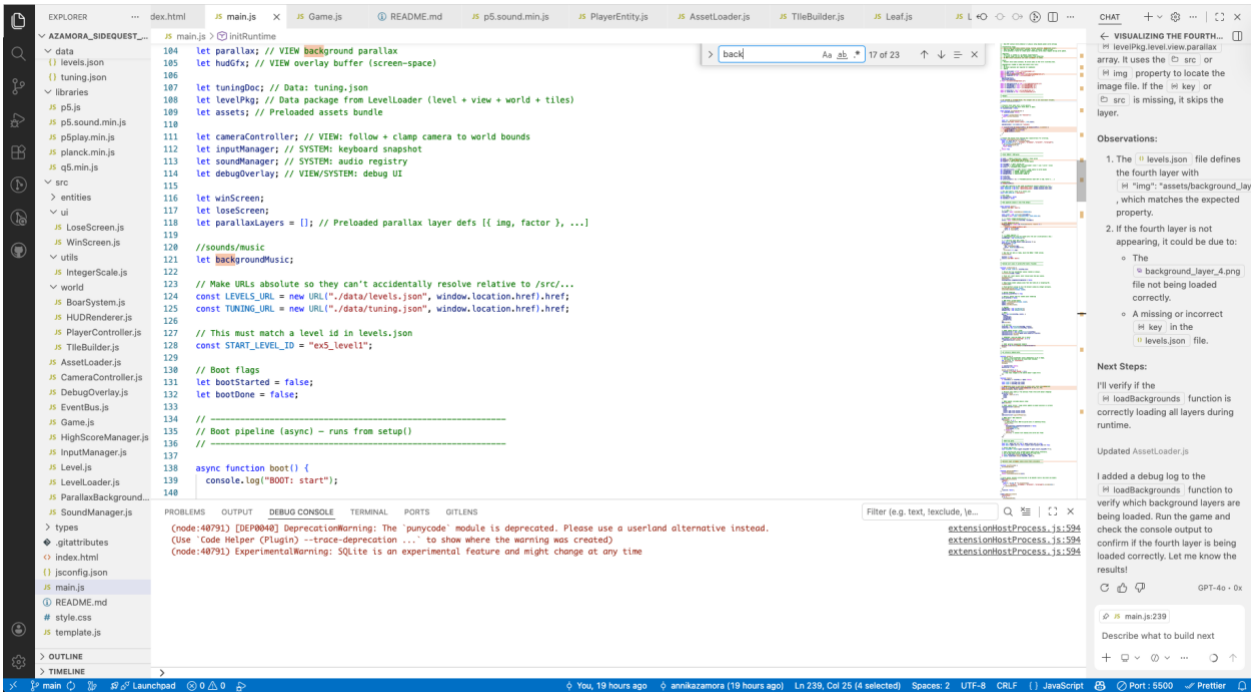
Side Quests and A4 (Individual Work)

- I was able change the background, tiles, player, and enemy, as well as add background music
- I was not able to add sound effects when the player jumped, collected an object, etc.
- I did attempt numerous times to add the sound effects, but my code kept breaking and I was on a time constraint
    - The overall quality of the game is still there, it just has less features than asked for

Role-Based Process Evidence



Drawing the bell sprite image



Adding background music

Goal of Work Session

- Modify example code 5 from LEARN to match the side quest requirements
  - Change player and enemy sprites
  - Change leaf image
  - Add background music
  - Add sound effects based on player interaction

Tools, Resources, or Inputs Used

- Visual Studio Code Copilot Add-on
- ChatGPT Auto
- LEARN Week 6 Example 5 Code

GenAI Documentation

If GenAI was used (keep each response as brief as possible):

Date Used: March 8, 2026

Tool Disclosure: Visual Studio Code Copilot Add-on, ChatGPT Auto

Purpose of Use: To change/add to the example code to meet side quest requirements

Summary of Interaction: I used the Visual Studio Code Copilot add on to ask where certain parts of the code were located since there were a lot of files and lots of code. I also asked it to include a fourth layer for the background since the background design I downloaded had four layers whereas the example code only had three. I did end up using ChatGPT though because the Copilot add on wasn't very reliable in the sense that it would either break my code, or say it was doing things that weren't displaying on the screen properly. I found that ChatGPT worked a lot better so I ended up sticking with that GenAI tool for the rest of the side quest.

Human Decision Point(s): Any time the code stopped working after a modification from the Copilot tool, I rejected the changes, re-evaluated and tried to identify what changes occurred and why it didn't work (missing bracket? wrong order?). Then I either tried for an alternative prompt or completely scrapped that idea.

Integrity & Verification Note: I ran the code after every change to make sure it still worked which also made it a bit easier to troubleshoot since I would know what changes had been made before the code stopped working.

Scope of GenAI Use: The main area that I did without the use of GenAI was changing the sprites/images. I found them online, named and uploaded each of the files, then I modified some code that initially called for the previous file name, to the current one.

Limitations or Misfires: I find that even in previous uses, Copilot would say it did what was prompted but the code would either be broken or not display properly. I started off using it, but after I prompted to add background music, it made the code crash, which I then decided to try using ChatGPT. The only limitation about ChatGPT is that although I can upload my code files, it's more difficult to translate it's corrections into my code since it cannot directly edit it. This made the process take a bit longer, but in my opinion, it was a reasonable trade-off since I much preferred the output of ChatGPT compared to Copilot on Visual Studio Code.

Summary of Process (Human + Tool)

- Tested/ran the code after each modification
- Approached each section step-by-step so that I wouldn't feel overwhelmed, and so that the prompts could be more detailed and area-focused
- Told the GenAI tools what changes I made, then prompted it to further process that area of code
    - Ex, I uploaded the music file, named it, then asked it to write code to load and play the background music

Decision Points & Trade-offs

- I couldn't get the sound effects to work for player movement/interaction
    - I worked at it for a couple of hours and decided to leave it due to time constraints

    - It was very frustrating but I realized there were other good components of the code that I got to work
    - I am going to try to figure it out, even after submitting the side quest, since sound effects could be an aspect of our group project that we would like to add
- Another key decision I made during this process was switching to ChatGPT
- I wanted to try something new since I was getting frustrated with the outputs from Copilot on VS code
    - It definitely wasn't the most efficient process, however, I found the output to be much better compared to Copilot

Verification & Judgement

- Ran/tested each modification after I or the GenAI tool modified something
- Referenced the starter example code
    - Ex, the code was running very slowly, so I referenced back to the original example code to see if it was different (it was). I then had to find out what was slowing my code down/why it was lagging
- Referenced the side quest rubric/requirements

Limitations, Dead Ends, or Open Questions

- I could get the sound effects to work, even with the use of both GenAI tools
- For submission and deadline purposes, I had to leave it be without, however I plan to come back and work through it
- When changing the leaf sprite to the bell, I initially wanted to rename all variables and naming conventions in the code so that it would make sense
    - However, it was difficult to do so even with GenAI
    - I had to restart my code after trying to change all of the naming conventions
    - I decided to keep everything as "leaf", and I only had to change the file name it was looking for since I named the file I uploaded "bell.png"

Appendix

Input: where do the layers of the background load in the file
Output: The layers of the background are loaded in the loadBackgrounds function in AssetLoader.js:188-213. This function dynamically loads background layers based on the parallaxLayers property from the levelPkg object. If parallaxLayers is not provided, it falls back to default background layers:

bgFar from background_layer_1.png

bgMid from background_layer_2.png

bgFore from background_layer_3.png

The function ensures that each layer is loaded asynchronously using loadImageAsync.

Input: load in a fourth layer for the background
Output: A fourth background layer (bgExtra) has been added to the loadBackgrounds function. It will load from background_layer_4.png by default. Let me know if you need further adjustments!

Input: the fourth layer isnt visible
Output: The ParallaxBackground class in ParallaxBackground.js:1-80 is responsible for rendering the background layers. It uses the layers array passed to its constructor, where each layer has an image (img) and a parallax factor (factor).

To ensure the fourth layer is visible, you need to verify:

The bgExtra layer is included in the layers array when constructing the ParallaxBackground instance.

The factor for bgExtra is set appropriately to determine its movement speed relative to the camera.

Let me check where the ParallaxBackground is instantiated to confirm if bgExtra is being passed.

The ParallaxBackground instance is created in the initRuntime function in main.js:187-236. It uses the parallaxLayers array, which is constructed earlier in the boot function.

To ensure the fourth layer (bgExtra) is visible:

Verify that parallaxLayers includes an entry for bgExtra with its image and a factor value.

Check that the draw method in ParallaxBackground is correctly iterating over all layers, including bgExtra.

Let me confirm how parallaxLayers is populated in the boot function.

Read main.js, lines 117 to 154

The parallaxLayers array is populated in the boot function in main.js:117-154. It uses the levelPkg.level.view.parallax property to define the layers. Each layer in defs is mapped to an object with:

img: Loaded using loadImage(d.img).

factor: A numeric value derived from d.speed.

To ensure the fourth layer (bgExtra) is visible:

Confirm that levelPkg.level.view.parallax includes an entry for bgExtra with its img and speed.

Verify that the loadImage call for bgExtra succeeds.

If bgExtra is missing from levelPkg.level.view.parallax, it won't be included in parallaxLayers. Let me check the levels.json file to confirm its contents.

The levels.json file does not include a fourth layer (bgExtra) in the parallax array under view. Currently, it only defines three layers with their respective images and speeds.

To make the fourth layer visible:

Add an entry for the fourth layer in the parallax array, such as:

Ensure the background_layer_4.png file exists.

Would you like me to update the levels.json file to include this fourth layer?

Input: yes
Output: The fourth parallax layer has been added to levels.json with the image background_layer_4.png and a speed of 0.8. Ensure the image file exists in the assets directory. Let me know if you need further assistance!

Input: I uploaded the file for background music called "backgroundMusic.mp3". Load the music and play it so that it loops
Output: The background music file backgroundMusic.mp3 has been successfully loaded and set to loop in the game. The changes are error-free. Let me know if you need further assistance!

Input: why is the level not working
Output: The levels.json file appears to be correctly structured, and the main.js file does not show any errors. However, the issue might be related to the background_layer_4.png file or how the parallax layers are being loaded.

Let me verify if the loadAssets function in AssetLoader.js is correctly handling the parallax layers.

The loadAssets function dynamically loads the parallax layers using the loadBackgrounds function. If the fourth layer is not appearing, the issue might be with the loadBackgrounds implementation or the background_layer_4.png file.

Let me inspect the loadBackgrounds function to ensure it handles all layers correctly.

The loadBackgrounds function dynamically loads the parallax layers from the levelPkg.level.view.parallax array. It uses the src or img property to locate the image file. If the key or src is missing, it skips the layer.

Observations:

The levels.json file defines the fourth layer with "img": "assets/background_layer_4.png", which matches the expected property.

If the fourth layer is not appearing, it could be due to:

The background_layer_4.png file not being loaded correctly.

A missing or incorrect key in the levels.json file.

Next Steps:

I'll verify if the loadBackgrounds function is correctly loading all layers during runtime.

I added a debug log to the loadBackgrounds function to verify which background layers are being loaded. Run the game and check the console output to confirm if the fourth layer is being loaded correctly. Let me know the results!

ChatGPT Conversation: https://chatgpt.com/share/69ade96f-50fc-800c-bdf2-b44ad68bbed4